using this calculated corneal power to determine IOL power. The idea of a "fudge factor" is certainly both desirable and practical. There are, however, several points about this study that need further scrutiny. The authors make the assumption that using the "historical method" leads to the most accurate corneal power. This is the gold standard by which all the other four methods are compared. In the historical method, as described by the authors, the change in spherical equivalent induced by LASIK at the corneal plane is subtracted from post-LASIK keratometry. In doing so, the authors do not take into consideration that there is not a one-to-one relationship between corneal power and refractive power. In other words, when the refraction at the corneal plane is changed by 1 D, the readings do not change by the same amount.

We draw attention to a similar published study that addresses this issue.[2] We performed a study of IOL calculation after both myopic and hyperopic LASIKs and arrived at similar conclusions. Our data verified a linear relationship between the amount of LASIK correction and the IOL power error. In addition, we also found that to compensate for this error after myopic LASIK, the pre-LASIK IOL power had to be increased by a factor of 0.7 for every diopter of myopic correction induced by LASIK. In the same study, the use of the historical method as described by Hamed et al[1] was also analyzed. Our results indicated that although the use of the method increased accuracy, after LASIK for myopia of more than 2 D, this method still underestimated the IOL power.

What exactly is the source of error in determining true corneal power after keratorefractive surgery? The answer most commonly cited is the alteration in the relationship between the anterior and posterior corneal surface that results in a change in refractive index. One may ask whether excimer ablation of the cornea inherently changes the corneal stroma, thereby partially altering the refractive index. There may be some evidence supporting this. A study by Patel et al[3] demonstrated that different layers of the cornea have slightly different refractive indices. Perhaps selective removal of layers of the cornea by the excimer laser results in a change in refractive index. In Hamed and colleagues' study,[1] when using the Gaussian formula, the authors employed a refractive index of 1.376 for the anterior surface of the cornea. However, this value is true only of a native cornea and not for an ablated cornea after selective removal of particular layers. This may partially explain why this approach, though taking corneal thickness into account, may still result in errors.

In summary, the authors' conclusions and recommendations appear to verify the results of previously published studies.[2,4] The most accurate method of calculating IOL power at this point remains use of refractive change induced by LASIK and a fudge factor to compensate for the change in refractive index of the cornea. In doing so, the relationship between corneal power and refraction needs to be considered.

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Author reply

Dear Editor:

We enjoyed reading Drs. Feiz and Mannis' letter and welcome the study of new approaches for calculating intraocular lens (IOL) power in patients who have undergone LASIK and photorefractive keratectomy. However, we do not feel that their letter accurately characterizes their method or the current state of knowledge regarding this challenging problem.

In their critique of the historical (more commonly referred to as "clinical history") method, the authors state that "there is not a one-to-one relationship between corneal power and refractive power." This misses the point. The clinical history method is based on the obvious fact that the refractive change induced by the excimer laser procedure occurs in the cornea. Therefore, the change in refractive error is simply vertexed back to the corneal plane, which is where this change took place.

We are surprised by their comment that the clinical history method "still underestimated the IOL power." As they state in their article, none of these methods has been tested in a clinical series. In their method, they divide the LASIK-induced refractive change by 0.7 and add this to the IOL power that was calculated for the pre-LASIK eye. A potential flaw in this approach is that it could result in overestimation of postoperative IOL power, resulting in unanticipated postoperative myopia. Therefore, any comparative evaluation of these methods is purely speculation, pending further clinical study.

In their critique of our use of the Gaussian formula, they comment that the refractive index of 1.376 is not likely to be accurate. We disagree, because 1.376 represents the refractive index of the air–tear interface and should not be affected by changes in refractive indices of various layers of the cornea.

Finally, we point out that a unique advantage of the method we presented in our article (adjusting postoperative corneal measurements according to the amount of LASIK-induced refractive change) is that it requires only one datum from before LASIK—the amount of myopia. This can often be provided by patients even though they cannot obtain copies of preoperative keratometry or topography measurements, which are required for both the clinical history and Drs. Feiz and Mannis' methods.

We look forward to seeing the results of applying these various methods in an actual clinical series. More importantly, as we indicate in our article, the ultimate goal is the development of accurate methods of measuring true corneal power.

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Dear Editor:

I offer some insights regarding the data presented by Hamed et al[1] that should allow the application of this information for all cataract surgeons in a practical manner.

Although the clinical history method of determining the true central corneal refractive power was found to be the most accurate way of determining the true refractive power of the cornea,[1–3] this method requires that the keratometry values before the LASIK procedure be available. Since the time between the LASIK and cataract procedures will often extend to more than 10 years and the LASIK and the cataract surgery will often be performed by different surgeons, pre-LASIK keratometry will be difficult and sometimes impossible to obtain for many patients.

The contact lens method has also been proposed as an accurate way of determining the true corneal refractive power.[1–3] This technique is cumbersome to perform, which makes practical application of this technique in a busy cataract practice difficult. It is my experience that this technique is seldom used.

Computerized videokeratography can also be used to estimate the corneal refractive power; however, Hamed et al[1] have demonstrated that this will produce misleading values, as it uses a standardized corneal refractive index that is known to have been changed by the refractive surgery. Although adjustments in this value may increase the accuracy of this value,[4] the variety of corneal topography machines available will lead to confusion regarding the correct adjustment factors for each machine.

Based on the results of Hamed et al and my personal experience with this problem over the last 8 years, I offer the following recommendations for the adjustment of the keratometry value and hence the intraocular lens (IOL) power after LASIK:

1. The IOL power is calculated in the standard manner utilizing the keratometry values and the axial length.
2. The amount of LASIK refractive correction is obtained from the patient or medical records. This value is usually easier to obtain, as patients may be aware of their previous correction or have an old pair of glasses, or old LASIK records may be available.
3. Subtract 1 diopter from the mean measured readings for every 4 D of myopic LASIK correction. This is based on the assumption that the true central keratometry value will be approximately 75% less than the measured keratometry value. This percentage reflects the results of previous studies indicating that the 3.0-mm keratometry values underestimate the true refractive change at the cornea by 24%[5] and therefore indicate only 76% of the induced corneal flattening.
4. Aiming for a postoperative refractive goal of −0.50 to −1.0 D would introduce another margin of safety to avoid postoperative hyperopia but would also increase the LASIK enhancement rate. Postoperative myopia is a much preferred postoperative outcome as it will be better tolerated by the patients and, if necessary, much more easily corrected by a LASIK enhancement.
5. Patients should be counseled preoperatively that a small LASIK enhancement may be required after the cataract procedure.

Although these recommendations involve estimations, they should provide an easily applied practical method that all cataract surgeons can use with their existing equipment and formulas to achieve excellent postoperative cataract refractive results in most cases while avoiding the highly undesirable postoperative hyperopia surprise.

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Author reply

Dear Editor:

We thank Drs. Probst and Holladay for their comments. However, we are somewhat confused by their recommendations. In point 1, they recommend calculating intraocular lens power in the standard manner using the measured keratometry values. Then, in point 3, they recommend using readings that have been modified by subtracting 1.0 D for every 4.0 D of myopic LASIK correction. In fact, this latter recommendation is nearly identical to what we have recommended in our article, in which we suggest that readings be reduced by 0.24 D for every diopter of myopic LASIK correction. Therefore, for 4.0 D of LASIK correction, the keratometry reading should be reduced by 0.96 D.

Unfortunately, as we point out in our article, using keratometry readings is not as accurate as using the EffRP value that we determine from the EyeSys device (EyeSys Vision, Houston, Texas). Therefore, we hold to our recommendation to use the EffRP values whenever they are available. The keratometry values can be used if the EyeSys device is not available for measuring the patient.